

GAURAV BANGADE

Software Engineer | Platform Architecture, Full-Stack & Offline-First Systems

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SUMMARY

Software Engineer at a maritime SaaS company since November 2024, with ~1,300 commits and ~1,180 pull requests across 15+ repositories. Founded and architected CVMS, a plugin-based, offline-first ship–shore platform, and created Ensign, the company's design system. One of the main frontend authors of Ecosailer, a production emissions-compliance product. Strongest in React, TypeScript, NestJS and Django REST Framework.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL
- **Frontend:** React 18/19, TanStack Router, Query, Table & Form, Redux, Zustand, Formik, Zod, ECharts, Recharts, Tailwind CSS, Vite
- **Backend:** NestJS, Fastify, Prisma, Django REST Framework, Django Channels, pg-boss, Node.js, Express
- **Databases:** PostgreSQL, MySQL, Redis, MongoDB
- **Distributed Systems:** Offline-first sync, transactional outbox, hybrid logical clocks, WebSockets, fault simulation
- **Security:** Role-based access control (RBAC), privilege ceilings, httpOnly cookie auth, JWT, Argon2id, XSS sanitisation, security audits
- **Testing & Quality:** Vitest, Playwright, React Testing Library, MSW, ESLint, Husky, Docker, Git
- **Design:** Design systems, design tokens, accessible data palettes, responsive UI
- **Maritime Domain:** IMO CII, EU MRV / ETS, FuelEU, IMO DCS, bunkering and ROB reporting
- **AI / LLM:** LangChain, RAG, FAISS, Model Context Protocol (MCP)

PROFESSIONAL EXPERIENCE

Software Engineer | ecoSAIL Infotech, Pune, India | Nov 2024 – Present

- **CVMS (in development):** Founded and architected a plugin-based, offline-first ship–shore platform on NestJS, Fastify, Prisma and PostgreSQL, with lint-enforced module boundaries, a transactional outbox, a hybrid logical clock and a simulation arena that tests sync under 10 satellite-link profiles and clock skew.
- **Ensign Design System:** Created a 26-section, token-driven design system with light and dark themes; it is the only styling system allowed in CVMS and the channel-partner portal.
- **Ecosailer:** One of the main frontend authors of the production emissions-compliance product (960+ merged PRs on its UI): built the Voyage Estimator, reporting wizards, analytics dashboards and the trial-period feature across 9 release lines.
- **Voyage Estimator:** Built a two-step wizard estimating a voyage's CO2, CII rating and EU ETS allowances, moving every regulatory constant into database masters so regulation changes need data, not deploys; covered by 55 tests.
- **Access Control & Security:** Rebuilt RBAC with a privilege ceiling and deny-by-default route manifest, moved auth to httpOnly cookies, added DOMPurify XSS sanitisation and ran a security audit across all 4 company repositories.
- **Ecosail Global & Helpdesk:** Built the shore hub's UI from scratch (React 19, TanStack Query, Zustand) and designed its helpdesk: ticket lifecycles, webhooks, machine-to-machine API keys and real-time WebSocket notifications.
- **Code Quality & Releases:** Removed every "any" type from the UI in a 6-phase, lint-locked migration, cut polling requests 10x and closed 12 critical and 18 high-priority defects to ship release 2.4.0.

PROJECTS

- **InquireAI** (Python, LangChain, FAISS, Streamlit, MongoDB): RAG pipeline that answers plain-language questions grounded in uploaded PDFs.
- **Artisan Studio** (React, Node.js, Express, MongoDB, Cloudinary): full-stack marketplace with artisan profiles, product uploads, reviews and buyer messaging.
- **JSON Toolkit** (React, TypeScript, React Flow): web tool to format, validate and explore JSON as an interactive graph.

EDUCATION

Master of Computer Applications (MCA) | MES' Institute of Management & Career Courses, Savitribai Phule Pune University | SGPA 8.0 | 2024

Bachelor of Computer Applications (BCA) | Kamla Nehru Mahavidyalaya, Nagpur | 84% | 2022

LANGUAGES

English, Hindi, Marathi